ITDBADM Machine Project (MP): Final Project Guidelines

1. Project Overview

Create a fully functional **Online Store Database Application** with a connected Graphical User Interface (GUI). The project demonstrates your ability to design, administer, and optimize a relational database while integrating it with a front-end application.

Task	Due Date	Submission
Database Proposal	**Nov 3 (Mon)**	Upload the final proposal via AnimoSpace
Final Project Deliverables	**Nov 17 (Mon)**	Upload all files via AnimoSpace
Project Demo	**Nov 17 – Dec 1**	Live demo during scheduled presentation (Class Time). Note: Actual demo. No recordings!

2. Proposal Guidelines

The proposal should be in Google Docs and include the following:

- **Store Name**: A unique and creative name for your online store.
- **Product/Service Description**: Brief explanation of what your store sells (e.g., gadgets, clothing, digital services).
- **Target Customers**: Identify your main market (e.g., students, professionals, gamers).
- **Key Features**: List the planned features of the online store (e.g., multi-currency support, order tracking).
- **Team Members & Roles**: Identify each member and their roles (e.g., frontend, backend, database admin).
- **Generative AI Declaration**: Indicate if generative AI tools were used for proposal proofreading or grammar checking (allowed).
    - **Allowed**:
        - Proofreading and improving grammar or writing style of the proposal or report.
        - Identifying *possible* transactions, reports, or features as brainstorming support.
        - Light "vibe coding" or code-style suggestions **for reference only**, provided the group can explain and rewrite the code themselves.
    - **Not Allowed**:
        - Direct generation of the database model/ER diagram.
        - Direct generation or copy-pasting of production-level SQL scripts, stored procedures, triggers, or GUI code.
        - Any AI-produced code or design that is not fully understood and re-implemented by the team

3. System Requirements

Database

- Minimum of **6 tables**, including:
  - **Users**, **Products**, **Orders**, **Order_Items**, **Currencies**, **Transaction_Log** (additional tables encouraged).
  - **Should have at least 2 branches and 3 currencies.**
- At least **10 stored procedures and 10 triggers** combined.
- Use **Virtual Machines** to create users and grant privileges.

GUI Front-End

- Programming Language: **Java**, **Python**, **PHP, etc.** (your choice).
- Minimum Features:
  - **Login** with at least **3 roles** (Admin, Staff, Customer)
  - **Browse Products/Services** (at least 6 products or services)
  - **Currency Handling** (support at least 3 currencies: PHP, USD, KRW)
  - **Admin Panel** for adding/editing products/services
  - **Add to Cart and Place Order (Transaction Management)**
  - **View Order History**

4. Deliverables (to be submitted on Nov 17)

1. **SQL Files**
   - Database schema (tables, keys, relationships)
   - Stored procedures and triggers
   - Transaction/backup scripts
2. **GUI Source Code**
   - Complete front-end application files with database connection
3. **Presentation Slides (PPT)**
   - Online store introduction
   - ER Diagram
   - Description of stored procedures and triggers
   - Screenshots of the GUI
   - SQL code snippets and sample outputs

5. Presentation & Demo

- **Duration:** 15-minute live demo + 10-minute Q&A
- Show the **working GUI** integrated with the MySQL database.
- Highlight stored procedures, triggers, transaction handling, and privilege management.

6. Additional Notes

- Use any IDE (e.g., Eclipse, PyCharm, VS Code) as long as it connects to MySQL Workbench.
- The Virtual Machine should be integrated and part of the GUI integration rubrics.
- Ensure your application runs smoothly during the demo. Incomplete or non-functional systems will affect grading.

7. Rubrics for assessment:

CRITERIA	Exemplary	Satisfactory	Developing	Beginning	Unacceptable
Grade	**4**	**3**	**2**	**1**	**0**
Database Design & Normalization	Database design clearly exceeds the minimum with additional well-normalized tables, enhanced relationships, or advanced business logic	At least 100% of the required core tables are correctly normalized and fully related with PK/FK currencies that reflect proper business rules.	Only 70–99% of the required tables are normalized, or branches/currencies are incomplete	Only 40–69% of the required tables are normalized, or branches/currencies are largely missing or incorrectly related.	Less than 40% of the required tables are normalized, or branches/currencies are missing altogether.
Stored Procedures Implementation	Stored procedures are fully functional, well-documented, and go beyond the minimum in quality, efficiency, or complexity.	At least 100% of the required stored procedures (baseline ≈ 10) are functional and support core transactions.	Only 70–99% of the required procedures are implemented or contain logic errors.	Only 40–69% of the required procedures are implemented or mostly non-functional.	Less than 40% of the required procedures are implemented or functional.
Trigger Implementation	Triggers are fully functional, well-documented, and exceed the minimum in quality, efficiency, or complexity.	At least 100% of the required triggers (baseline ≈ 10) are functional and handle validations/logging.	Only 70–99% of the required triggers are implemented or contain noticeable errors.	Only 40–69% of the required triggers are implemented or are mostly non-functional.	Less than 40% of the required triggers are implemented or functional.
Transaction Management & ACID Compliance	Transactions and logging exceed minimum expectations, demonstrating robust ACID compliance and advanced features.	Minimum transactions are implemented with proper COMMIT/ROLL BACK and accurate logging to meet ACID properties.	About 70–99% of required transactions demonstrate ACID compliance but with some inconsistencies.	About 40–69% of required transactions demonstrate ACID compliance or logs are incomplete.	Less than 40% of required transactions demonstrate ACID compliance or logs are missing.

CRITERIA	Exemplary	Satisfactory	Developing	Beginning	Unacceptable
Grade	**4**	**3**	**2**	**1**	**0**
User Privilege Manageme nt	Role-based access and security measures exceed the minimum through finer-grained permissions or advanced configuration s.	At least 100% of the required roles (baseline ≈ 3 roles: Admin, Staff, Customer) are implemented using GRANT/REVOK E with correct permissions.	Only 70–99% of the required roles or permissions are correctly implemented.	Only 40–69% of the required roles or permissions are correctly implemented.	Less than 40% of the required roles or permissions are correctly implemented.
GUI Design & Usability	GUI is visually appealing, intuitive, and surpasses the minimum with professional layout, responsive design, or enhanced user experience.	GUI meets all minimum functional and design requirements: clean layout, clear navigation, functional login, product browsing, ordering, and currency handling. The Virtual Machine is integrated.	GUI connects to database but layout is cluttered or confusing in at least two areas or only 70–99% of required features work.	GUI is poorly organized or difficult to navigate with only 40–69% of required features functional.	GUI lacks required features or is not usable.
Integration, Presentatio n & Delivery	Database and GUI integrate seamlessly with zero errors; presentation slides are polished and complete. presenters speak confidently, explain technical details clearly, and engage the audience professionall y.	GUI is integrated with database; presentation includes ERD, SQL snippets, screenshots, and a working demo with clear explanations; presenters deliver the report and demo clearly and audibly.	Integration is inconsistent or demo contains noticeable errors; presentation lacks clarity or only 70–99% of required components included; delivery shows limited confidence or clarity.	Integration is unstable and presentation is poorly organized with only 40–69% of required components included; delivery is hesitant or difficult to follow.	No working integration or presentation submitted; students unable to explain the project.
				TOTAL	**28.0**